

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE)

November 2013

CE 208: Object oriented Concepts and Visual C++ Programming

Date: 15.11.2013, Friday

Time: 1:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION - I

- Q - 1 Answer following questions:** [07]
1. What is Class? How it is differ from Object? 02
 2. Explain concept of Polymorphism in C++ with example. 01
 3. What is Constructor? Explain concept of default constructor? 02
 4. What is purpose of Virtual class? 01
 5. Explain concept of Static member of class. 01

- Q - 2 Attempt following :** [14]
- (a) Explain public, private and protected access specifiers and show their visibility when they are inherited as public, private and protected. 07
 - (b) Explain friend function with example and list the special properties of friend function. Also explain merits and demerits of friend function. 07

OR

- Q - 2 Attempt following :** [14]
- (a) Write a VC++ program to create a Class Student that has Id_no, Name, City and Result as Data member and Create void Get_Students(void) and void Show_Students(Id_no) as member function. 05
 - (b) What do you mean by type conversion? Give an example of basic to object conversion. 04
 - (c) Write a C++ program demonstrating use of the pure virtual function with the use of base and derived classes. 05

- Q - 3 Attempt following :** [14]
- (a) Define operator overloading? Explain how to overload unary operator and binary operator. 05
 - (b) How to use constructor in VC++? Explain constructor overloading with example. 05
 - (c) Briefly explain Windows program structure. 04

OR

- Q - 3 Attempt following :** [14]
- (a) Which operators cannot be overloaded? Write steps to overload + operator so that it can add two complex numbers. Explain with example how can a function template be created. 05
 - (b) What MFC? Explain document/view architecture in detail. 05
 - (c) Briefly discuss about different data types with examples. 04

SECTION – II

- Q - 4 Attempt following :** [07]
1. What is purpose of Abstract class? Explain syntax for defining Abstract Class. 02
 2. What is difference between Class diagram and Object diagram? 02
 3. Which symbol is used for Aggregation relation and Composition relation in Class diagram? Give example. 01
 4. List features of Object Oriented Programming. 02

- Q - 5 Attempt following :** [14]
- (a) Explain function overloading with a C++ program to perform addition. 04
 - (b) Explain in detail about multiple inheritance with a detailed example program. 04
 - (c) Define a class Person with a set of data members and member methods of your own. Write a C++ program to write and read the instance of user-defined class Person to and from a file. 06

OR

- Q - 5 Attempt following :** [14]
- (a) Draw Class diagram for ACCOUNTING SYSTEM that keep record of following information: Name of Depositor, Account no., Type of A/c, Balance Amount. Which can perform operations like: To assign initial values, deposit amount, withdraw after checking balance, To display name and balance. 06
 - (b) Differentiate the following: 04
 1. Activity and State
 2. Generalization and Specialization
 - (c) "Encapsulation and polymorphism improve reusability". State this statement is TRUE or FALSE. Give justification for your answer in detail. 04

- Q - 6 Attempt following :** [14]
- (a) Explain various symbols of Activity diagram. Draw Activity diagram of any system with explanations. 07
 - (b) What is State Diagram? Draw State Diagram for ATM example and explain various symbol used in it. 07

OR

- Q - 6 Attempt following :** [14]
- (a) What is Sequence diagram? How it is differ from Collaboration Diagram? Explain with example. 07
 - (b) What is Usecase? Draw Usecase diagram of ATM example and explain various symbols used in it. 07
